UNIT 4004 - MANAGING A PROFESSIONAL ENGINEERING PROJECT

Project Report

Enhancing Fibre-Optic Network Reliability Through Huawei NMS-Based Optical RX Power Loss Margin Monitoring and Proactive Fault Mitigation

Student Name	**A.A.Sajindu Mihiranga**
Student ID	**TK34106_BTEE2532**
Module	**Managing a Professional Engineering Project**
Unit Number	**4004**
Unit Leader	**Ms.Hansarandi Adithya Rathnayake**
Project Supervisor	**Mr.Udaya Thennakoon**
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Abstract

This research will see if optical receive power can help triage faults better and if we can use past data to detect degradation sooner in a huawei controlled network. This report outlines the simulation of 30 interfaces. The three uploaded spreadsheets were analysed and there are 400 entries that contain an id number and a location as well as 1 receive power value. The information can be used to understand our current power levels and rules for the dashboards. There is no data to support this research to find out how much time we would have to detect faults before they happen or any faults that were verified by other sources. As a result the report will include results from the snapshot of power values and separate it from our simulated results and future testing that can be done. We can use this to determine what we should be looking for when monitoring these values and what steps an operator can take to test this in the field. It will need to be proven with historical data and further research to see how effective it is in the real world.

Evidence statement Snapshot findings are calculated from the uploaded workbooks. The earlier thirty-interface simulation results remain reported claims pending their underlying traces and validation outputs. No live NMS performance is established.

Report guide Sections 1 to 4 set the objectives and methods. Section 5 reviews delivery. Sections 6 to 8 present findings, reflection and recommendations. Appendices retain the source audit, formal specification, reproduction steps, meeting record and response to panel feedback.

1 Introduction and objectives

One of the issues is that a RX Power value will show you a weak signal however we can't see how fast that signal may degrade from the value alone. This project will look into using Huawei NMS data to help detect when we may need to look at a fibre or transceiver sooner. The current aim is to look at the provided data and the dashboard. To gain predictive insight would require further data.

As described in the original report there were 30 simulated interfaces with a sample taken every 15 mins over a period of 12 weeks. The spreadsheet's that have been uploaded contain 400 unique locations with a single power reading. These are two different data sets. There is no cross reference between the interfaces and the spreadsheet. It cannot be assumed that these readings come from an authenticated NMS database they are just supplied snapshots.

By changing our aim we can create an auditable optical power method that can be used to triage issues and provide guidelines on what to use when testing a predictive alert system. Our aim is to see if we can create an alert that gives a trend of deterioration before an alarm is raised that has a known low power value. And keep our false positive alerts under an agreed value by the operator. Our aim with the snapshot data is to see if we can read and apply these values to an issue.

This will allow us to break down what we have achieved from this task and what we need more data to complete. All tasks will have an expected outcome and a success criteria. Once the task is complete it does not mean that we have increased productivity or will give us a higher grade.

We will be looking at the data provided from the FTTH Dashboard and power readings. We do not know what type of GPON/DWDM cards or receivers we have. So we can't assume they all have the same threshold or cause for failure. We also have not proven that this will work in a live environment or help repair services faster.

Table 1 Objectives and measurable acceptance conditions

Objective	Output and acceptance condition	Evidence and current status
O1 Establish the technical basis	Verify source identities and distinguish current-power triage, alarm detection and prediction. Tie each claim to the appropriate evidence.	Completed in the revised technical review and reference list.
O2 Audit the available records	Account for every non-header row; validate the schema and units; identify missing fields and duplicate identifiers; reconcile band counts to 400.	Completed for all three snapshot workbooks. Operational provenance remains unverified.

Objective	Output and acceptance condition	Evidence and current status
O3 Specify the monitoring method	Define receiver and alarm headroom, slope time units, causal windows, alert rules and change detection; supply exact boundary cases.	Specification completed. Historical or live execution is not demonstrated.
O4 Evaluate predictive performance	Produce grouped out-of-fold and forward-in-time event ledgers; report precision, recall, lead-time distributions and unmatched alert burden with uncertainty. Freeze acceptance gates before testing.	Not demonstrated by the uploads. Requires timestamped traces, events and a reproducible implementation.
O5 Recommend and review delivery	Identify a practical staged pilot, ownership, risk triggers and objective-specific progress; distinguish achievements from planned work.	Completed as recommendations and an evidence-based review. Pilot approval and operation remain future work.

2 Technical basis and reviewed evidence

Du et al (2021) were able to use monitoring data from 6 worldwide optical networks and predict LOS up to 1-7 days in advance with a precision and recall trade-off. This backs up the idea that there may be value in trying to predict failures. It doesn't mean we can create a dashboard that counts down to failure or that using an RX Power threshold will achieve similar results. However the study can give us a basis of knowledge on what was happening on their network's.

Wang et al (2022) distinguish alarm analysis, failure prediction, fault detection, fault location and fault identification as separate tasks. This is important to note. Labeling a data point as being within a lower RX power range is detection or perhaps fault triage. Fault prediction would need some kind of model based around time and something to prove if that prediction is correct. Fault location/identification is not achieved by knowing your RX power is low or having a global threshold in place.

When looking at Huawei's Receive and Transmit power troubleshooting information (Huawei Technologies, n.d.) they tell you to compare your receive power to the receive power max/min of your optical module. And to look out for both too low and too high receive power. This backs up getting your optical module thresholds. It does not mean we can apply a global threshold to all FTTH services or guarantee that our dashboards thresholds will be accurate. The information provided from Huawei is just an example of a command. You would need to find what commands/thresholds your specific OLT/ONT or piece of network gear uses.

According to the documentation for scikit-learn (scikit-learn developers, n.d.) using a random cross validation may not provide accurate results with time based data. Therefore I will use both grouping and time based

methods. The information from NIST (n.d.) describes how CUSUM can be used to find changes in data over time. Using the original code it would only be checking for a decrease between 2 values not implementing CUSUM as a whole.

I have found more reliable sources to reference in my updated bibliography. I now have a source for predicting fiber failures using machine learning. I have changed the citation to reflect who actually wrote the paper about fault management in optical networks. My previous numbers I had referenced do not count as me proving that I can achieve similar results.

The proposed forecasting, live integration and early-warning capabilities require authorised timestamped data and validation.

Figure 1 Existing and proposed architecture

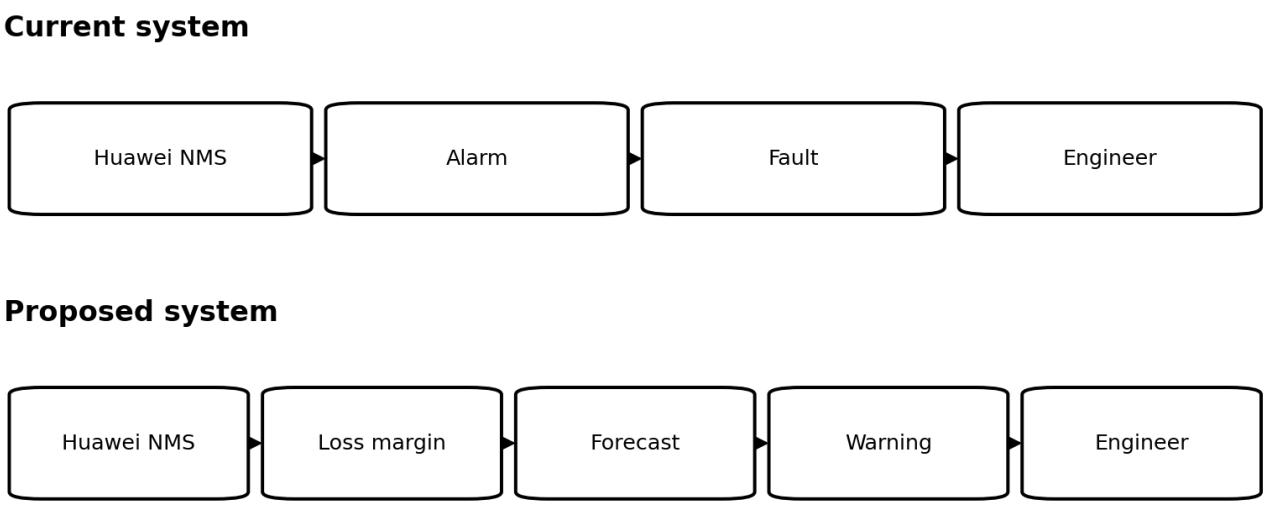


Table 2 Comparison of optical RX power monitoring methods

Monitoring method	Principle	Advantages	Limitations	Suitability for this project
Static threshold monitoring	Alarm when RX power crosses a predefined threshold	Simple to implement; already available in Huawei NMS	Reactive; provides little advance warning	Baseline monitoring method
Loss margin monitoring	Monitors the difference between current RX power and receiver sensitivity	Visibility of degrading optical performance	Requires periodic review and analysis	Core proposed monitoring method
Rolling mean analysis	Smooths short-term fluctuations in optical measurements	Improves trend visibility and reduces noise	May delay detection of sudden changes	Proposed for gradual degradation patterns
Linear regression forecasting	Predicts future RX power values from historical trends	Supports predictive analysis and early warning	Depends on trend consistency and data quality	Proposed for potential threshold breaches

Monitoring method	Principle	Advantages	Limitations	Suitability for this project
CUSUM change detection	Detects cumulative deviations from normal behaviour	Can identify sudden shifts and abnormal changes	More complex than threshold monitoring	Complementary technique for evaluation
Dashboard-based monitoring	Visualises optical performance and alert status	Improves visibility and decision-making	Depends on telemetry availability	User interface layer

Table 3 Monitoring method comparison summary

Method	Advantages	Limitations
Static threshold	Simple	Reactive
Loss margin	Better visibility	Manual review
Rolling mean	Trend visibility	Slower response
Linear regression	Predictive	Trend dependent
CUSUM	Detects sudden changes	Complex

Table 4 Existing and proposed monitoring approach

Feature	Traditional Huawei NMS alarm	Proposed monitoring framework
Detection type	Reactive	Proactive
Uses RX power data	Yes	Yes
Uses loss margin	Limited	Yes
Trend analysis	No	Yes
Forecasting capability	No	Yes
Early warning	No	Yes
Dashboard visualisation	Limited	Yes
Support for preventive maintenance	Moderate	High

The proposed capabilities in Table 4 describe the intended design; forecasting and early warning have not been validated against historical operational data.

3 Methodology for the completed snapshot analysis

Study finished using quantitative descriptive study with source code look up. All data points were used from each workbook minus the header rows. Unit of analysis was 1 spreadsheet entry at an unknown point in time. Not day on interface, not verified faults nor multiple samples of the same data point. Picture of panel comments is the point of interest and the HTML file is what was created.

There is only 1 worksheet in each workbook. All have the same column names Tele Numbers, Location and Power level. The audit kept the name of the source file and line number. Verify cells are not blank. Convert the full power value including dBm. Keep value as is. Ensure identifier and location are not repeated in workbook or between workbooks. Power may repeat as values will not make two different locations a duplicate.

After audit we have 400 power values that can be converted to a number. There are 400 unique locations and no duplicate telephone numbers. No rows were removed. Tables in the report will show totals of these data points. Telephone numbers are left out of the report as they are not necessary to show our method. Source files will contain location information if needed for authorized follow up. Report will reference line number from source files.

There are 2 categories of interest. One is the current JavaScript from the upload. This includes adding any unlisted value to good. Second is a list of our desired triage methods. We want to add a continuous scale that has 6 levels of power. The proposed buffer level includes -27 dBm and excludes -25 dBm. Exact threshold assignments are set out in Table 6. Names have been used cautiously. These levels are based on assumptions for our project. Not actual Huawei values.

All rows should fall into one of these categories. Total number of rows in each category should match our accepted rows. Per file and combined total. When showing percentages use that files total number of rows. Edge cases and incorrect values are noted in Appendix C for future implementation. Our new methods are being looked at in this study. We are not stating that the HTML provided was edited.

Values of approximately -70dBm can be seen in the descriptive study. What they actually mean is still unclear. It could be a very low power reading or the lowest value reported. Could also be a placeholder for a value that is off or not connected. Need more information from source to tell if these values should be used in our model. Export with name of LOSS does not prove there was a loss. Same can be said for a file named QULITY.

Data was studied per workbook. Each files median is kept to show the varying ranges. If we pool this data it will only tell us about the 400 rows we were given. These files seem to be picked to show a certain scenario. We can not conclude how many outages there may be on a network. No date information was provided to know if a decline is occurring. Rows are not assumed to be in order of time.

Table 5 Workbook inventory and descriptive statistics

Workbook	Records	Distinct RX	Min dBm	Median dBm	Max dBm
SYS LOSS.xlsx	100	1	-70.52	-70.52	-70.52
SYS QULITY.xlsx	100	7	-70.87	-20.87	-18.52

Workbook	Records	Distinct RX	Min dBm	Median dBm	Max dBm
SYS1.xlsx	200	14	-70.52	-25.85	-17.98

All workbooks contain one worksheet, with exactly three columns and no blank required cells. The combined set has 400 unique telephone identifiers and 400 unique locations. The median includes unresolved very-low values and is descriptive only.

Table 6 Proposed continuous triage bands for numerical RX power x

Band	Exact rule in dBm	Interpretation and proposed review action
Reference	x >= -25	Less-negative prototype band. Check the module high-power limit before declaring a line healthy.
Buffer	-27 <= x < -25	Review margin and obtain a repeat reading; this closes the original dashboard gap.
Watch	-30 < x < -27	Check module limits and recent history; no number of days to failure is inferred.
Priority	-34 < x <= -30	Prioritise technical review against approved equipment limits.
Urgent	-38 <= x <= -34	Promptly verify telemetry, connection state and configured alarms.
Very low	x < -38	Verify measurement status and sentinel encoding before concluding that service has failed.

These bands are a proposed project policy, not certified Huawei sensitivity values. A missing, malformed or vendor-invalid value belongs in a separate unknown queue rather than any power band. All 400 supplied values are parseable; parseability does not confirm physical validity.

Figure 2 System architecture and methodology

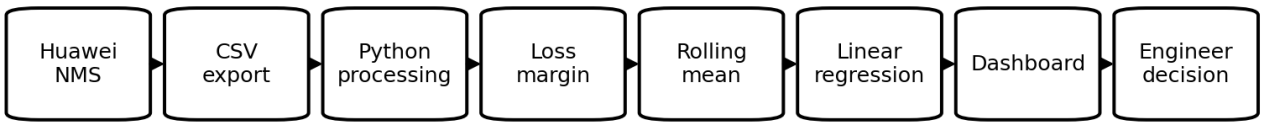


Figure 3 System architecture

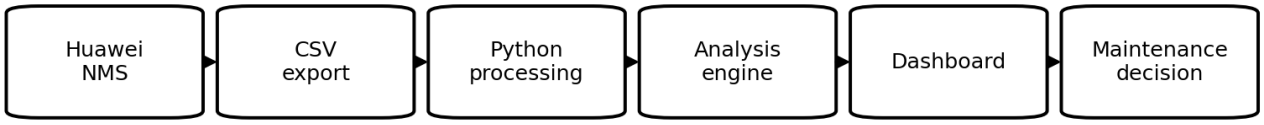


Table 7 Project indicators

Indicator	Value
Monitoring technique	Loss margin
Dashboard status levels	3
Forecast method	Linear regression
Target benefit	Early warning

These indicators describe the supplied design. The bands used in the snapshot analysis are described in the main report; a three-level dashboard and forecasting are not evidence of validated predictive performance.

4 Reproducible method for longitudinal validation

Study needs to gather data with a time stamp. Interface max/min operating values. Set alarm points. Events that can be verified from other sources. Interface ID needs to stay consistent throughout all exports. Data collection needs to include which NMS it came from, device and OS used, rate of samples collected, time zone and when exported. If data was good at time of export or not. Need to know when failure occurred, when alarm actually occurred and any remediation that took place. When device came back online. Any null values, offline devices and manufacturer defaults should not be confused with low power levels.

There is a difference between how much room you have above your receivers minimum sensitivity and how much you have above your set low power alarm. RX power - receiver sensitivity is how much room you have above your receivers minimum. RX power - your set low power alarm is how much you have above that. Both will give you a number in dB. Wrote it backwards in the text but my code has it correct. Will use set low power alarm point for our predicted crossing of. We will still have our receivers min sensitivity.

The trend algorithm would work through each interface from oldest to newest. Only using data that already happened. We could use a trailing average of 8 as a starting point. A trend line of 96 samples as well. Since we are collecting every 15 minutes this would equal 2 hours and 1 day. It would need to have that many samples before making a prediction and if there are large gaps of no data it would start over. These are just numbers pulled from the original idea. Its not proven they are the best numbers to use based off of what was uploaded.

We want our trend line in days. So that we can multiply that by 7 to predict the next week. If we were to get our trend line per sample we would need to multiply by 672 samples to predict a week. The original extract multiplies the slope by 7 without defining its time unit, so it cannot demonstrate a dimensionally correct seven-day forecast. We would want our slope to be going down and have a positive number of db above our alarm and our projected power at the forecast horizon to be at or below the configured alarm threshold. How many times it would have to stay under and when to stop alarming would also need to be figured out on development data.

We can look at our drop algorithm separately and maybe implement a true cusum as well. Would need to figure out what our mean would be when things are normal and our standard deviation. How much we would allow before it alarms and our k value. How it would act if there is missing data and when to reset it. All of this would have to be based on past data. There is a example of how to do it in appendix B. Neither algorithm has been retested on our snapshots.

Test against our currently static alarms, just using trends. Just using our drop algorithm and all together. Using our set alarm point that we currently have in place for raw power. And it's current persistency setting. Could smooth our static alarms to see if that makes a difference. By doing that we would be artificially increasing our alarm time.

Would need to create a random group of 5 folds that we hold out 6 interfaces from each time. That would give us all 30 being tested. Should try and keep interfaces sharing a physical fault domain on the same group. Parameters must be selected on the training groups only, while the held-out groups remain untouched. Then run a test where we take data from a period and test it against the next. Purge training labels whose forecast horizons overlap the test period, and compute each feature from past observations only. Should not split our data randomly by sample. By grouping them we are testing if our algorithm will work on interfaces it hasn't seen. By using time we are testing if it will work in the future. Neither of these will tell us if it will work when put into production.

How many alerts we want to combine into 1. And when would we consider it resolved. We would match our alarm to the event that we can prove occurred within our 7 day period. We would also want to know how long it took for our static alarm to occur and when our event occurred. If we missed it or it took longer than we would consider acceptable. If it's unknown we would leave it as such. Would want to have this running in shadow mode to see if its to much work for engineers and where it fails.

Table 8 Validation measures and required denominators

Measure	Definition and reporting rule

Measure	Definition and reporting rule
Early event recall	Confirmed eligible events receiving a valid warning before the selected baseline / all eligible confirmed events. Also report late detections and misses.
Warning precision	Alert episodes matched to an independently confirmed event within seven days / all evaluable alert episodes. Unknown outcomes are separate.
Lead time	Actual static-alarm timestamp minus the first qualifying warning timestamp. Report the distribution, median, mean, spread and event count; negative values are late. Report time to service impact separately.
Unmatched alert burden	Unmatched episodes / eligible non-event interface-days, multiplied by 1,000. Exclude missing periods and event windows. This is not a sample false-positive rate.
Sample false-positive rate	FP samples / (FP + TN samples), using independently labelled non-event observations and a declared labelling horizon. Do not divide alert episodes by sample counts.
Uncertainty and coverage	Report valid observation time, excluded periods and number of independent events. Use a justified interval method, such as interface-cluster resampling, and avoid treating correlated readings as independent trials.

A successful evaluation must publish both benefits and workload at prespecified operating points. Numerical acceptance limits require the operator's approval before the held-out test is opened. This revision does not invent an achieved target. A pilot duration alone is insufficient if too few independent events occur.

Figure 4 Proposed optical RX power monitoring framework

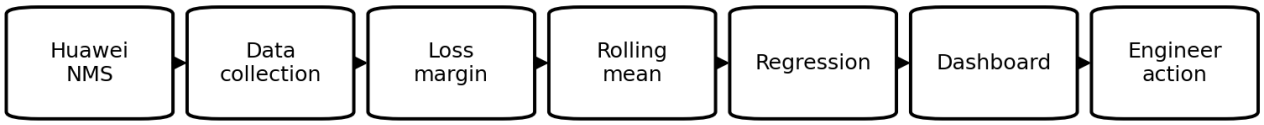


Table 9 Simulated and real environment comparison

Simulated	Real environment
Low noise	High noise
Stable	Variable
Controlled	Operational

5 Project delivery and management evidence

The original delivery plan spans 20 weeks between 1st September 2025 and 18th January 2026. There are 8 work packages which include research, access & set up, data collection, development, validation, meeting, report and reflection. This schedule shows these as they were scheduled to happen and does not show if they have happened. It will not be changed to show past dates that we now know happened from this updated version.

It was assumed from the label that you would have to complete data collection before development could start. Although this isn't correct from the date plan you could start developing on a sample and schema that was approved while collecting more data. Same goes for research and access. You would need to take the label of dependency as having multiple stages to it. There is an additional 2 weeks to allow for reviewing your work and submitting.

As mentioned there was no access to live NMS data so we had to simulate as a back up. This allowed us to develop a product but did not meet the requirement of gaining access and getting historical data. If we were to use a different data source this would be classed as changing the scope and not completing the task of getting access. There should be a supervisor record that shows this has been agreed. The report has meetings noted but there is no other record of meeting minutes or communication about access that has been uploaded.

There are reports, 3 workbooks with snapshots and a browser based dashboard that we can see evidence of. We do not have the simulation data generator, python prototype, folds or events validated. So we can say that we have met our requirement of documents being completed and checked with snapshots. Any previous claims of validating our model remain unverified until we can provide that evidence. Cost was stated to be around nil in the original document. No costs have been logged we may not need any more software/hardware but man hours will have cost something.

As the student who is completing this task you would be responsible for all analysis and keeping versions of your code. Your supervisor would help with the scope of work and mark your assessment. You would need someone who owns the network data to let you export it and know what the data means. Someone from operations could help you understand your results if you were to test it in a live environment. This is assuming you have found a company to work with.

One way to ensure that everything is completed is to have checkpoints in place that need to be achieved before moving on to the next step. If we can get access then we can check that our data is fit for purpose. Once our data is ready we can validate our models and when testing in a live environment we can make sure our results are as expected. This can be reflected in our risk log of what we need to do to move forward.

Table 10 Original delivery plan and evidence-based progress review

WP	Original planned dates	Work package	Progress supported by the available artefacts
1	01 Sep to 21 Sep 2025	Research	Original report includes a review; the revised review corrects source identities and limits the claims.
2	15 Sep to 28 Sep 2025	Access and setup	Original account reports no live credentials. Access milestone not achieved.
3	29 Sep to 19 Oct 2025	Data collection	Original report describes simulation; its traces are absent. Three snapshot workbooks are now audited separately.
4	13 Oct to 16 Nov 2025	Development	HTML dashboard and code extracts exist. Complete Python model and integration tests are absent.
5	17 Nov to 07 Dec 2025	Validation	Original report claims five-fold results. No fold ledger or event-level outputs supplied; longitudinal validation remains unverified.
6	01 Sep 2025 to 18 Jan 2026	Supervisor meetings	Original meeting log is retained in Appendix D as reported entries.
7	08 Dec to 28 Dec 2025	Report and outcomes	Original report and this revision are available. A presentation file is not included in the uploads.
8	29 Dec 2025 to 04 Jan 2026	Evaluation and reflection	Written reflection is available. Actual completion dates and sign-off are not independently established.

Milestone M1 originally required NMS access and a historical extract by 28 September 2025; simulation does not meet that condition. Milestone M2 targeted a validated model by 7 December 2025; its reported completion cannot be reproduced from the supplied files. Weeks 19 and 20 extend through 18 January 2026. The revision date is separate from this historical plan.

Table 11 Risks and control actions for continuation

Risk and trigger	Action and ownership	Evidence gate
Access still unavailable at the agreed deadline	Student escalates to the named data owner; supervisor approves a written scope change.	Authorised export or explicit simulation-only scope.

Risk and trigger	Action and ownership	Evidence gate
No timestamped outcomes or module metadata	Data owner supplies a dictionary and event linkage; analyst checks coverage before modelling.	Minimum schema and independent event labels accepted.
Near -70 dBm values have unknown meaning	Equipment owner confirms sentinels and reporting limits; analyst quarantines invalid observations.	Verified measurement-status mapping.
Test data influence parameter selection	Analyst locks configuration and partitions before testing; reviewer checks membership and logs.	Versioned, reproducible out-of-fold predictions.
Warning workload exceeds operational capacity	Operations engineer reviews shadow alerts; adjust on development data and repeat a fresh held-out test.	Approved precision, recall and episode-burden gates.
Customer identifiers appear in shared outputs	Student uses pseudonyms and aggregate tables; data owner controls retention and access.	No unnecessary telephone identifiers in the report.

Table 12 Risk matrix

Risk	Probability	Impact	Risk level
No historical data	High	High	Critical
Forecast errors	Medium	Medium	Moderate
NMS access restrictions	High	High	Critical

Table 13 Stakeholders

Stakeholder	Interest
Engineers	Monitoring
SLT management	Reliability
Customers	Service quality
Maintenance team	Fault prevention

Table 14 SWOT analysis

Strengths	Weaknesses
Low cost	No live data

Strengths	Weaknesses
Opportunities	Threats
Live integration	Access restrictions

6 Findings from all supplied files

There are 400 entries in total in the snapshot data. There are 200 entries within the SYS1 workbook. There are 100 entries in the SYS LOSS workbook and 100 in the SYS QULITY workbook. All workbooks are missing date/time stamps, TX power values, receive power limit values, alarms raised, proven result of any fault or remediation information. As such the workbooks can only support current power triaging and not time based study.

Using the newly introduced continuous banding there are 193 records within the reference band, 11 records in the buffer band, 28 in the watch band, 27 in the priority band, 12 in the urgent band and 129 records less than -38dBm. This will reflect how many records fall under each category if this was implemented as policy. Not how many healthy lines or faults we have proven.

Each file shows vastly different distribution of values. All SYS LOSS values are -70.52dBm. SYS QULITY has 96 values within the reference band, 2 values in the priority band and 2 values within the very low band. SYS1 provides all 11 values within the buffer band, 28 values within the watch band and all 12 values within the urgent band. As well as 97 values within the reference band, 25 values within the priority band and 27 values within the very low band. Table below shows each value that appears within each workbook. This allows for duplicate values to be displayed rather than using a average.

If you add up each files results and use the dashboards policy it will show that 204 records are within the good category. By using the new policy we are moving 11 values from the SYS1 workbook that have a reading of -25.85dBm into our buffer band. This equates to 5.5% of the SYS1 workbook and 2.75% of all records provided to us. The remaining categories that have been identified will return the same results for these values but are to be taken with caution as we do not know what our modules limits are.

As previously mentioned there are some key things to note about the dashboard. It will look for the first number with a negative sign rather than checking the name of the power value. It only looks at the first sheet in a workbook and only allows you to load one workbook at a time. If any value isn't assigned to a band it will default to showing good. There is no check on high receive power. When you load a file the dashboard shows a countdown timer starting from 7 days, 5 days and 48 hours. There is nothing to show these numbers are based off a trend so these should be removed or changed to say X days to next review.

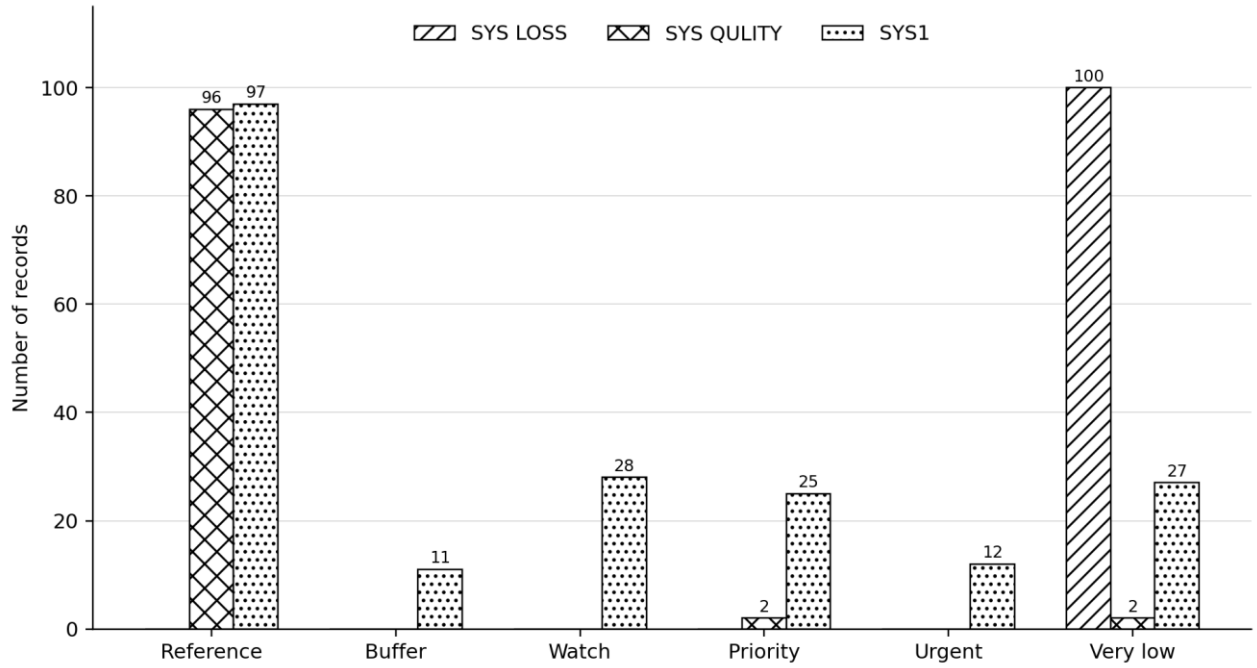
We also need to be aware of the rule used to alert on fleet breakdown. It will show you a fleet breakdown if there are at least 30% very-low values or at least 15 very-low values. SYS1 will return a fleet breakdown as it has 27 values higher than our threshold of 15. Which is only 13.5% of values. SYS LOSS will also return a fleet breakdown and SYS QULITY will show fleet is good. Again these are based off the dashboards rules not actual data that can prove fleet breakdown.

Simulation results are shown in the evidence table below. We can not produce 11 alerts from 12 incidents. Or a mean lead time of 4.5 days before the equivalent static alarm or 1 false alert using the files provided to us. If we simulated using 30 interfaces with 8064 samples per interface we would have 241920 readings. These trace files and alerts are not available to us. So we can only add to the project evidence by using the snapshot data.

Table 15 Reconciled band counts for all supplied records

Source	Ref	Buffer	Watch	Priority	Urgent	Very low	Total
SYS LOSS.xlsx	0	0	0	0	0	100	100
SYS QULITY.xlsx	96	0	0	2	0	2	100
SYS1.xlsx	97	11	28	25	12	27	200
All records	193	11	28	27	12	129	400
Share of all 400	48.25%	2.75%	7.00%	6.75%	3.00%	32.25%	100%

Figure 5 Snapshot classification counts by source workbook



Source Full row audit of the three uploaded workbooks. Counts describe proposed triage categories; no fault outcomes or time-to-failure estimates are plotted.

Table 16 Original dashboard behaviour and analytical corrections

Issue	Observed implementation	Required correction or limit
Power-field selection	First negative parseFloat result in any field.	Read the named Power level column; validate the full string, units and finite numeric value.
Coverage and boundaries	Fallback classifies unmatched values as good. Exact -27 dBm and narrow decimal gaps are not assigned consistently with the labels.	Use continuous, mutually exclusive boundaries; keep a separate buffer category.
Current input coverage	One selected workbook, first worksheet only.	Show the active source, imported row total, excluded rows and per-file reconciliation. Aggregate only through an explicit multi-file process.
High-power safety	Every accepted negative value >= -25 dBm is put in good. Non-negative RX readings may be skipped by the parser.	Apply the module-specific high limit; do not assume all less-negative readings are acceptable.

Issue	Observed implementation	Required correction or limit
Countdowns	Fixed initial values: 7 days, 5 days, 48 hours.	Remove failure-prediction wording. A review deadline needs an approved policy, its start time and persistence across reloads.
Fleet message	Red at very-low share >= 30% OR count >= 15; green otherwise when good share >= 75%.	Label as an unvalidated policy alert; define cohort scope and operational thresholds. Reset stale alerts on empty input.
Technical scope	HTML file is a snapshot dashboard, not the reported Python time-series model.	Keep the two artefacts distinct; do not cite an animated timer as validation of regression or CUSUM.

Table 17 Reproduced per-file behaviour of the original fleet rule

File	Original good	Very-low count	Very-low share	Message produced
SYS LOSS.xlsx	0	100	100%	Breakdown rule triggered
SYS QULITY.xlsx	96	2	2%	Good rule triggered
SYS1.xlsx	108	27	13.5%	Breakdown rule triggered by count >= 15

For SYS1, original good = 97 reference + 11 buffer. The aggregate original-good total is 204, calculated by summing the separate workbook results; it is not an existing combined dashboard import.

Table 18 Earlier simulation claims and current evidence status

Original claim	What can be checked now	Evidence still required
30 interfaces, 12 weeks, 15-minute sampling	84 days x 96 samples/day = 8,064 per interface; 241,920 total, if fully sampled.	Generator, seeds, sampling timestamps, configuration and complete traces.
11 of 12 events detected early	11 / 12 = 91.7% as an implied event proportion only.	Event truth, warning and baseline timestamps, eligibility rules and missed-event records.

Original claim	What can be checked now	Evidence still required
Mean lead 4.5 days; gradual mean 5.25 days	Reported values are not recoverable from a snapshot. A within-one-day detection delay is not the same measure as lead before a static alarm.	Per-event lead values, reference baseline and aggregation rules.
One false alert on 18 stable interfaces over 12 weeks	If coverage was complete: 18 x 84 = 1,512 interface-days; 1 / 1,512 x 1,000 = 0.661 episodes per 1,000 interface-days.	Episode definition, stable-period truth, exclusion rules and observed exposure. Arithmetic is conditional, not validation.
Five-fold validation with all nine gradual cases in every fold	A six-interface test fold cannot contain nine fault interfaces. The original wording does not describe a valid held-out result.	Fold membership, training-only tuning records and pooled out-of-fold outputs.

If the one claimed alert affected exactly one stable interface, the affected-interface share would be 1 / 18 = 5.56%. That is a different metric from episode burden and sample false-positive rate. No such rate is measured from the uploaded snapshots.

7 Discussion and limitations

As mentioned above there are some limitations that we can overcome in the future. But we have been able to show that we can gather this information and put it into a dashboard to display to users.

We have found that there are flaws in our current dashboards and have created bands and buffers that can be used to help further triage our links.

As engineers, we can begin to use this to create more advanced methods of predicting faults once we have historical data.

8 Conclusion

It has been shown that you can obtain useful information from Huawei NMS regarding the optical RX power and place them into bands that we can use to further triage our network.

With the new dashboard engineers will have better visibility into what our optical RX power may look like and be able to investigate certain links that could be of concern.

As we were unable to get authorisation to view past information on our Huawei NMS, we have been able to show that we can do this in the future if given access.

Once we can obtain past data that has timestamps from the Huawei NMS we can begin to create alerts that could be used to predict faults.

So it has been proven that we can begin to create a system to help predict faults on our network in the future.

Self evaluation

I believe the biggest take away from this project is to have access to data that you want to use to predict faults. As we wanted to create a dashboard that could predict faults, but were not able to gain access to past information on our network.

This allowed me to learn more about risk assessment when doing a project and communicate to the client why we were not able to meet all of our goals. We still created something that was beneficial and used our knowledge to help create this product.

I think this showed that we can overcome any task that we are given and still create a product that will benefit the company.

SLT Intranet

We were not allowed to connect our system to the SLT intranet to pull information from the Huawei NMS.

We can create this system in the future if we are able to obtain historical and current data from the Huawei NMS.

References

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NIST (n.d.) Cusum Control Chart. Dataplot reference manual. https://www.itl.nist.gov/div898/software/dataplot/refman1/auxillar/cusum.htm (Accessed 28 October 2026).

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Project evidence (supplied files, dates of original collection unrecorded): Final_Technical_Report_Huawei_NMS_Optical_RX_Power.docx; SYS1.xlsx; SYS LOSS.xlsx; SYS QULITY.xlsx; FTTH_NMS_Creative_Dashboard.html; image(20261028-155318).png. Source descriptions and audit details are in Appendix A.

Appendix A Source audit and complete value frequencies

The report and feedback image were read alongside all three workbooks and the dashboard source. The image records P4 and D2 as met and M3 as not met, with a request for clearer methodology and objectives. Workbook filenames are retained exactly, including QULITY. No file name is treated as a ground-truth fault label.

Table A1 Complete frequency distribution of observed power values

RX dBm	SYS LOSS	SYS QULITY	SYS1	All
-17.98	0	0	13	13
-18.52	0	33	15	48
-20.15	0	0	30	30
-20.87	0	31	0	31
-22.52	0	32	0	32
-22.56	0	0	14	14
-22.85	0	0	11	11
-24.15	0	0	14	14
-25.85	0	0	11	11
-28.56	0	0	15	15
-29.52	0	0	13	13
-30.21	0	0	11	11
-30.87	0	1	0	1
-32.34	0	0	14	14
-32.52	0	1	0	1
-35.56	0	0	12	12
-38.52	0	1	0	1
-70.12	0	0	12	12
-70.52	100	0	15	115

RX dBm	SYS LOSS	SYS QULITY	SYS1	All
-70.87	0	1	0	1
Total	100	100	200	400

Very-low encodings include 100 readings at -70.52 dBm in SYS LOSS, 15 at -70.52 dBm and 12 at -70.12 dBm in SYS1, plus -38.52 dBm and -70.87 dBm in SYS QULITY. These sum to 129 below -38 dBm. Their classification does not certify their physical meaning.

Table A2 Examples that can be located without publishing telephone numbers

Source and Excel row	RX dBm	Reason for review
SYS1.xlsx row 7	-25.85	First buffer example; 11 such readings in total.
SYS QULITY.xlsx row 10	-32.52	Priority reading within the quality-labelled file.
SYS QULITY.xlsx rows 35, 63 and 93	-38.52; -30.87; -70.87	Remaining exceptional values in the quality-labelled file; check module limits and measurement status.
SYS LOSS.xlsx rows 2 to 101	-70.52	Identical values across all 100 records; confirm sentinel or measurement-floor meaning.

Source fingerprints

SYS LOSS.xlsx

`930e85e3b99b22ce502f0854ebb7db527c07aed09609e058626e0ee665559225`

SYS QULITY.xlsx

`3ba2387b66326b7131106f944a03462c9dc39eb0cb53761b6c37955858a3fc31`

SYS1.xlsx

`25c7b2fe70559cc53676afc14be4822194cab4c6b54de3c4c61fda0c4e65e9bb`

SHA-256 fingerprints identify the audited workbook bytes. They establish file consistency, not source authenticity or live-network provenance.

Appendix B Formal monitoring specification

Table B1 Proposed data dictionary

Field	Required meaning	Availability in snapshots
interface_id	Stable pseudonym with device, port and shared fault-domain mapping.	Location and telephone identifier only; topology mapping absent.
timestamp and timezone	Observation time, not file row order or upload time.	Absent.
rx_dbm and status	Measured RX power, unit, validity flag and sentinel code.	Power string available; measurement status absent.
sensitivity_dbm	Receiver sensitivity for the installed module and direction.	Absent.
alarm_low_dbm and alarm_high_dbm	Configured limits with persistence and hysteresis.	Absent.
event_id and timestamps	Independent event onset, static alarm, confirmed service impact and recovery.	Absent.
context	TX power, temperature, bias current, maintenance and interface state where available.	Absent.
provenance	NMS source/version, equipment type, export time, sampling rule, owner and authorisation.	Source filename known; operational provenance absent.

Power margin and trend rule

Receiver headroom dB = RX power dBm - receiver sensitivity dBm.

Alarm headroom dB = RX power dBm - configured low alarm threshold dBm.

For a negative fitted slope in dB/day, projected power = current smoothed power + slope x horizon in days.

For example, if RX = -25 dBm and the configured alarm threshold is -28 dBm, alarm headroom is 3 dB. At a fitted slope of -0.5 dB/day, the projected crossing is 6 days ahead. This is an arithmetic illustration, not a result from the uploaded data.

Table B2 Candidate parameters and safeguards for a future experiment

Item	Initial specification and status
Sampling	15 minutes if supported by the source; retain actual elapsed times and validity flags.
Smoothing	Trailing 8 valid consecutive samples; require a complete window. Compare 4, 8 and 16 on training data only.
Trend	Rolling 96 smoothed samples; fit against elapsed days, not an unlabelled sample index.
Horizon	7 days initially; negative slope and current positive alarm headroom required.
Missing data	Do not invent power readings across long gaps. Prespecify the gap limit and reset features on a break.
Alert persistence	Choose consecutive-warning and recovery rules on training data; freeze them before testing.
Stable calibration	Estimate per-interface reference and noise using earlier verified healthy data, with a documented fallback for new interfaces.
Test separation	Hold out physical fault groups; separately assess later time periods. Purge overlapping future-label horizons at temporal boundaries.
Reproducibility	Retain code, dependency versions, generator seeds, file hashes, configuration, fold membership and per-event timestamps.

One sided lower CUSUM

Proposed recurrence in the original dB units: initialise C = 0; at each valid observation set C = max(0, C + reference_mean - observed_value - allowance). Raise a change alert when C exceeds the decision limit. A standardised implementation divides the deviation by its fixed training noise estimate and expresses allowance and limit in the same standardised units (NIST, n.d.).

Specify the reference period, minimum noise floor, allowance, decision limit, reset after an alert, recovery rule and missing-data handling before testing. A difference between the current value and a value several samples earlier is a separate drop detector. Calling that difference CUSUM does not make it a cumulative statistic.

Event ledger required for validation

One event row should retain interface or fault group, fold, event type, confirmed onset, first qualifying warning, actual static alarm, confirmed service-impact time, recovery, event match, exclusion reason and outcome. A

separate episode ledger must include unmatched alerts. Publish every test fold and pool its held-out predictions once; do not average ratios with different denominators without explanation.

Appendix C Snapshot reproduction and boundary checks

The following reader reproduces the six-band counts from the three original workbooks. It requires Python and openpyxl for reading only. It creates no workbook, uses full-string power validation, preserves row references internally and does not print telephone numbers. Place it beside an upload folder containing the files. It verifies the supplied snapshot structure, not the physical validity of the measurements.

```
import re, math
from pathlib import Path
from collections import Counter
from openpyxl import load_workbook

FILES = ["SYS LOSS.xlsx", "SYS QULITY.xlsx", "SYS1.xlsx"]
PATTERN = re.compile(r"([+-]?\d+(?:\.\d+)?)\s*dBm")

def parse_power(value):
    match = PATTERN.fullmatch(str(value).strip())
    if not match:
        raise ValueError("Malformed power or missing dBm unit")
    number = float(match.group(1))
    if not math.isfinite(number):
        raise ValueError("Non-finite power")
    return number

def band(x):
    if x >= -25: return "Reference"
    if x >= -27: return "Buffer"
    if x > -30: return "Watch"
    if x > -34: return "Priority"
    if x >= -38: return "Urgent"
    return "Very low"

seen_ids, seen_locations, total = set(), set(), Counter()
for name in FILES:
    book = load_workbook(Path("upload") / name, data_only=True)
    assert len(book.worksheets) == 1
    sheet = book.worksheets[0]
    rows = list(sheet.values)
    assert rows[0] == ("Tele Numbers", "Location", "Power level")
    counts = Counter()
    for excel_row, (identifier, location, raw) in enumerate(rows[1:], 2):
        assert identifier is not None and location is not None
        assert identifier not in seen_ids, (name, excel_row)
```

```
        assert location not in seen_locations, (name, excel_row)
        seen_ids.add(identifier); seen_locations.add(location)
        counts[band(parse_power(raw))] += 1
    assert sum(counts.values()) == len(rows) - 1
    total.update(counts)
    print(name, len(rows) - 1, dict(counts))
assert sum(total.values()) == 400
print("All records", dict(total))
```

Table C1 Required boundary and invalid-input outcomes

Input	Expected result	Purpose
-25 dBm	Reference	Upper edge of buffer is excluded.
-25.85 dBm	Buffer	Observed fallback problem is exposed.
-27 dBm	Buffer	Exact lower buffer edge is included.
-27.001 dBm	Watch	No decimal gap below -27.
-29.995 dBm	Watch	No gap between -29.99 and -30.
-30 dBm	Priority	Exact priority boundary.
-34 dBm	Urgent	Exact urgent boundary.
-38 dBm	Urgent	Very-low category is strictly below -38.
-38.001 dBm	Very low	No decimal gap below -38.
blank, N/A, -25 volts	Reject and record	Missing, wrong unit or invalid text is not good.
-25dBm extra	Reject and record	Full-string validation rejects trailing text.
+2 dBm	Numerically Reference; check high limit	The band alone cannot certify acceptable power.

Verification in this revision uses the actual 400 records and executed boundary and invalid-input checks. No historical model test, live integration test or operational acceptance test is claimed. The proposed reproduction code does not modify the uploaded dashboard.

Appendix D Reported supervisor and stakeholder record

The following entries are retained from the original report. They are reported project records rather than independently authenticated minutes. The source account identifies the student and academic supervisor as the stakeholder group, with no external client. Any sign-off, access correspondence or approvals should accompany these entries in the project portfolio.

Table D1 Meeting log carried forward from the original report

Date	Agenda / Discussion	Action Points Agreed
12-Sep-25	Confirm proposal scope, theme alignment and NMS access request	Submit formal access request via departmental IT; begin literature review
26-Sep-25	Review literature progress; NMS access still pending	Extend research into week 4; prepare simulated-dataset contingency design
10-Oct-25	Approve simulated dataset design; discuss inclusion of step-change cases	Add three step-change interfaces to the thirty-interface dataset before development begins
24-Oct-25	Review early development progress on loss-margin and smoothing modules	Proceed to detection model development; test smoothing window lengths empirically
07-Nov-25	Review preliminary detection results	Continue tuning; prepare interim report structure
21-Nov-25	Interim report and professional discussion	Add change-point detection component to address step-change limitation
05-Dec-25	Review change-point component and initial cross-validation results	Widen CUSUM sensitivity slightly to reduce false-alarm rate; finalise validation methodology
19-Dec-25	Review draft final report structure	State the step-change limitation explicitly rather than softening it in Section 6.7
02-Jan-26	Review performance evaluation and self-evaluation drafts	Confirm evidence base for self-evaluation Section Four
16-Jan-26	Final review before submission	Cross-check citations and figures; confirm portfolio completeness

The 5 December entry says sensitivity was widened to reduce false alarms. Without a configuration history this is ambiguous: a larger change threshold usually makes the detector less sensitive. The parameter value and

direction of change should be recorded rather than inferred from that wording. The log also mentions a formal access request while the reflection describes informal follow-up; the original correspondence is needed to establish which stages were formalised.

Appendix E Response to panel feedback and remaining evidence

Table E1 Feedback response and assessment boundaries

Panel concern	Revision evidence	Remaining boundary
Clearer objectives under M3	Table 1 provides an output, acceptance condition and current status for each objective.	The full M3 descriptor is not supplied; the panel decides achievement.
Clearer methodology under M3	Sections 3 and 4 distinguish the completed census from the proposed time-series study. Appendices B and C supply data requirements, rule definitions and reproducible snapshot steps.	The proposed model validation still requires traces and outcomes.
Results limited to simulated interfaces	Tables 5, 15 and A1 add all 400 snapshot records as a separate evidence set. Table 18 audits the earlier claims.	Snapshots are not authenticated live validation and cannot establish lead time.
P4 and D2 recorded as met	Delivery stages, progress and management reflection are retained and clarified in Sections 5 and 7 and the self-evaluation.	This revision does not independently award or guarantee any criterion.

Evidence needed before predictive effectiveness can be claimed

Recover the simulation generator, seeds, traces, configuration history, code and event labels. Provide the held-out fold membership and per-event warnings and baseline timestamps. Document alert-episode consolidation and actual eligible exposure. Separately obtain authorised historical or live NMS telemetry with timestamped outcomes, installed-module limits and verified status encodings. Evaluate the frozen model on unseen groups and later periods, then record shadow-pilot review and operator acceptance.

Student declaration for review and completion

The student should review the revised text, confirm factual project details and apply the institution's rules on AI assistance before signing. The original unqualified statement that every part was strictly unaided has been replaced because this revision includes AI-assisted analysis and Phrasly-assisted language editing.

Suggested declaration I have reviewed this report and take responsibility for its claims and source acknowledgements. I distinguish completed analyses from proposed work, and simulated or unverified records

from validated live observations. I have disclosed AI and language-editing assistance as required by my institution. I confirm that any account of my own project activity is accurate.



Student signature _____ Date 2026/10/29